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RESEARCH-ARTICLE

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Path Planning of Mobile Robot Based on RRT in Rugged Terrain

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ABSTRACT

The[1] basic RRT algorithm is difficult to realize the path planning of 3D environment. Aiming at the 3D path planning problem of mobile robot in complex environment. An improved RRT algorithm is proposed. Firstly, the 3D environment model is established for the path planning of mobile robot under complex rugged terrain, and the task requirements are analyzed. Then, based on the basic RRT algorithm, a new node acceptance criterion is proposed for the particularity of 3D environment. This acceptance criterion of new node take into account the variation of terrain height, the smooth of path, and the kinematics constraints such as turning radius and pitch angle of the mobile robot. Furthermore, the dynamic p_g value design is carried out, and the process of improving the algorithm is given. At last, simulation results show the effectiveness of the proposed method. The generated path passes at low altitudes, avoiding peaks and rough terrain.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; *Planning and scheduling*; Robotic planning

KEYWORDS

Path planning, Mobile Robot, Rapidly-exploring Random Tree, heuristic algorithm

1 INTRODUCTION

Path planning is to plan the feasible path from the initial position to the target position for the mission equipment (unmanned aerial vehicle, mobile robot etc.) on the premise of meeting certain constraints according to the requirements of the task [1]. The resulting path is optimal under specified performance metrics. Path planning is an important part of autonomous robot control.

The existing path planning methods can be divided into mathematical programming method, intelligent optimization method, heuristic algorithm, artificial potential field method and so on.

RRT is a path planning algorithm based on spatial sampling proposed by S.M.LaValle [2]. RRT has powerful path searching capability, and has higher optimization performance than mixed integer linear programming, swarm intelligence and random roadmap [3]. Unlike the path planning method based on Voronoi graph, the RRT algorithm does not need to divide the task area beforehand. Literature research shows that RRT algorithm can obtain feasible path more quickly than A* algorithm [4].

Since the RRT algorithm was put forward, researchers have proposed a variety of improved methods [5–7]. However, the current research on RRT algorithm focuses on the binary task space. That is, the task environment is either a completely infeasible obstacle area, or a completely non-threatening feasible area, which doesn't completely correspond with the actual application environment of mobile robots. For example, the fieldwork robot needs to face the rugged terrain environment. When the mobile robot undertakes reconnaissance, surprise

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attack and other tasks, it is necessary to select the optimal path according to the topography and human threat of the ground. It is of great practical significance to apply the RRT algorithm to 3D terrain which can describe the real environment more accurately. The algorithms applied to 3D path planning mainly include HPSO algorithm[8] and improved ant colony algorithm [9].

Based on the basic RRT algorithm, according to the special requirements of path planning in 3D environment, an improved RRT algorithm is proposed in this paper. A new random node selection method and the acceptance principle of new nodes are designed, and the algorithm steps are given. Finally, the validity of the proposed method is verified by simulation experiments.

2 PROBLEM DESCRIPTION

2.1 Environment Modeling

The mobile robot moves in rugged terrain. So the environment cannot think of as a two-dimensional plane. In order to create a more appropriate robot work environment, a 3D terrain model is established by computer simulation. The height of the terrain is superimposed by the height of the reference and the height of the fluctuation.

$$H(x,y) = H_0 + Z(x,y) \tag{1}$$

In the formula, $H(x,y)$ is the actual height of the coordinate point (x,y). H_0 is the reference height of this area. $Z(x,y)$ is the fluctuation height which superimposed on the reference height. The simulated field terrain is generated by a random process using Gauss distribution. The average height of a task area can be chosen as the reference height. Digital maps are stored in grid form. For the height of non-grid points, the interpolation method is used to obtain the terrain height. The resulting digital map is shown in Fig. 1.

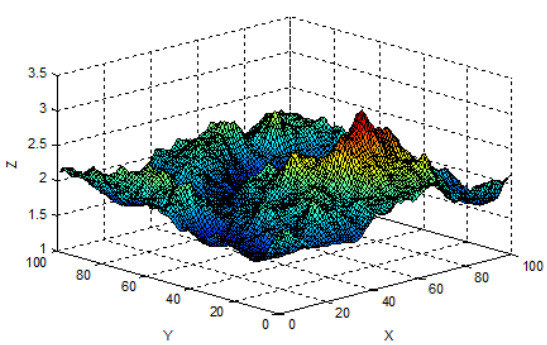

Figure 1: 3D terrain digital map.

2.2 Task Requirement

Suppose that the start position of path planning is S, and the goal position is G. The path is composed of multiple waypoints. The kth point is expressed as $S(x_k, y_k, z_k)$, where x_k, y_k, z_k are the coordinates and height of this position respectively. The adjacent waypoints form a path segment. A path consisting of K segments can be represented as:

$$P = \{S, S(x_1, y_1, z_1), \cdots, S(x_k, y_k, z_k), \cdots, G\} \tag{2}$$

For the path planning problem in 3D complex terrain environment, the path required by the planning algorithm meets the following requirements:

1） generating a feasible path from the start position to the target position;

2） The path should satisfy the dynamics and kinematic constraints of the mobile robot;

3） The path should tend to low altitude areas, such as valleys and saddle regions, to improve the safety of movement by through the smooth terrain height change.

The path should be as smooth as possible, avoid excessive steering and height change, and ensure the safety of robot.

3 ALGORITHM DESIGN

3.1 Basic RRT Algorithm

The basic RRT algorithm performs path planning in a two-dimensional environment. The path planning process of RRT algorithm is described in combination with Fig 2. First, the temporary target point q_{rand} is selected with probability p_g, and then the point closest to q_{rand} is selected as q_{near}. A new node q_{new} can be obtained by growing from q_{near} to q_{rand} with step ε. If there is no obstacle between q_{near} and q_{new}, q_{new} is accepted and an extension is completed. A feasible path can be obtained from the q_{init} to q_{goal} by extending enough times.

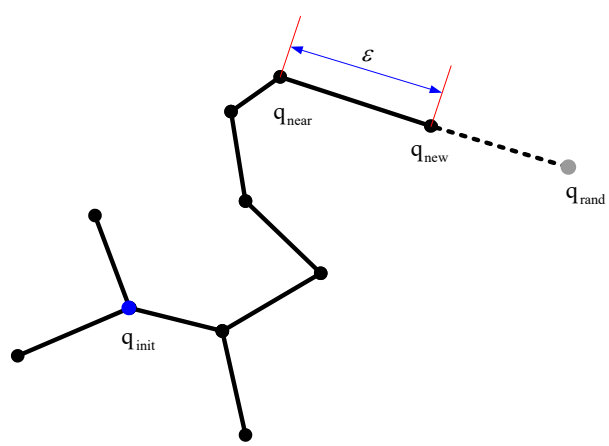

Figure 2: Nodes extension of the basic RRT.

RRT algorithm has probabilistic completeness, that is, as long as the target position and the starting position are in the same

connected region, then the nodes can be expanded from starting points to the target position and obtained a feasible path.

Aiming at the 3D terrain environment, the node sampling method and the node extension mode of RRT algorithm are modified respectively.

3.2 q_{new} Acceptance Criteria

For the binary task spaces with only feasible regions and unfeasible regions, traditional RRT algorithm can judge whether there is a threat between q_{near} and q_{new} as the receiving principle of q_{new}.

In the three-dimensional task environment, the new criteria of q_{new} acceptance is required to adapt to the continuous external threat, and at the same time to meet the robot performance constraints. The acceptance criteria of q_{new} include the following:

3.2.1 Design dead node. The robot is limited by the maximum moving distance. The path distance between the candidate new node q_{new} and the root node q_{init} is $L(q_{new}, q_{init})$.

$$L(q_{new}, q_{init}) < L_{max} \tag{3}$$

If $L(q_{new}, q_{init}) \geq L_{max}$, it means this path has exceeded the maximum moving range of the robot. Set the parent node of q_{new}, that is, the corresponding neighbor node q_{near} is a dead state, and the node is no longer considered when the node is expanded later. This method enables the resulting path to satisfy the range constraint.

3.2.2 Minimum turning radius. The turning radius of the robot must be large enough. Assuming that S_{i-1}, S_i and S_{i+1} are three adjacent nodes in a path. The turning radius at this three nodes is r_i, the constraint condition for S_{i+1} to be extended is:

$$r_i \geq r_{min} \tag{4}$$

r_{min} is the minimum turning radius of the robot.

3.2.3 Maximum pitch angle. The rate of variation of the robot's height is also limited by its mobility. The maximum pitch angle of robot moving is defined as θ_{max}, the height of two adjacent track nodes is z_k and z_{k+1} respectively, and the length of the track segment is l_k. The maximum pitch angle constraint can be expressed as:

$$\frac{|z_{k+1} - z_k|}{|l_k|} \leq tan\theta_{max} \tag{5}$$

3.2.4 Acceptance probability of q_{new}. If $h(q_{new}) \leq h(q_{near})$, that is, the height of the candidate new node is less than the height of the near node. This shows that the tree extends to low altitude area. Therefore, this q_{new} will be accepted

If only nodes with the height lower than q_{near} are accepted, RRT growth standstill may occur. Therefore, if the candidate q_{new} satisfy the condition of $h(q_{max}) \geq h(q_{new}) \geq h(q_{near})$, it will be accepted with the probability $p(q_{new})$.

$$p(q_{new}) = 1 - \frac{h(q_{new}) - h(q_{near})}{h(q_{max}) - h(q_{near})} \tag{6}$$

$h(q_{max})$ is the maximum height constraint that satisfies the pitch angle constraint.

$$h(q_{max}) = \varepsilon * tan\theta_{max} + h(q_{near}) \tag{7}$$

To sum up, the acceptance probability of q_{new} to be accepted probability p is expressed as:

$$p = \begin{cases} 1 & , h(q_{new}) < h(q_{near}) \\ 1 - \dfrac{h(q_{new}) - h(q_{near})}{h(q_{max}) - h(q_{near})} & , otherwise \\ 0 & , \{h(q_{new}) > h(q_{max})\} or \{L(q_{new}, q_{init}) \geq L_{max}\} \end{cases} \tag{8}$$

3.3 Dynamic p_g value

The main factor that affects the planning speed of RRT algorithm is the process of selecting q_{near} after obtaining q_{rand}, because this process must to traverse each node of the current random tree and calculate the distance from q_{rand} successively [10]. Therefore, the number of random nodes that need to be generated for each new node q_{new} is an important factor affecting the algorithm speed. In this paper, an adaptive parameter strategy is proposed to dynamically adjust the algorithm parameters p_g according to the N_r values. The value of p_g is as follow:

$$p_g = p_{gmin} + (p_{gmax} - p_{gmin}) * \frac{N_{rmax} - N_r}{N_r} \tag{9}$$

p_{gmax} and p_{gmin} are upper and lower limits of p_g respectively. If random nodes have been generated multiple times and still cannot extend a new node, the randomness of the algorithm should be enhanced to facilitate the random tree to expand to unknown areas. Therefore, as N_r increases, the value of P_g should be reduced gradually.

3.4 Improved RRT Algorithm Description

The improved RRT algorithm takes full account of the particularity of the 3D rugged terrain and the limitations of the robot itself. The probability value p_g can be dynamically adjusted according to the N_r value, so as to realize the randomness adjustment of RRT. The improved RRT algorithm flow is shown in Fig. 3.

The improved RRT algorithm can be described as follow:

1) Algorithm initialization;
2) Update p_g;
3) Whether to reach the goal position, not to arrive, turn to 4), else turn to 9);
4) Generate probability $p \in [0,1]$, if $p < p_g$ turns to 5), else turn to 6);
5) Select the target position as q_{rand}, determine q_{near}, and then get q_{new}, then turn to step 7);
6) Select a temporary target point q_{rand} in the blank space, and then get q_{new};
7) If q_{new} satisfies the acceptance criteria, then turn to 3), else turn to 8);
8) $N_r = N_r + 1$, turn to 2);

9) Find the path from q_{init} to q_{goal}.

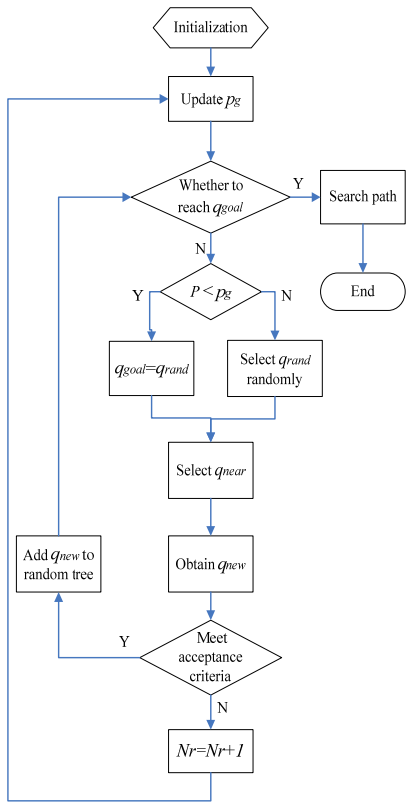

Figure 3: Flow chart of improved RRT algorithm.

4 SIMULATION

The created task environment is 3D rugged terrain. The starting position of the path planning is set to (5,95), and the target position is set to (100,100). The step size ε is 5, and the random node selection probability p_{gmin} and p_{gmax} are 0.4 and 0.7 respectively. Minimum turning radius is $r_{min} = 2$. The max pitch angle of robot is $\theta_{max} = 30^{\circ}$ and the range of robot is $L_{max} = 150$. The improved RRT algorithm is used to simulate the path planning.

MATLAB 2014a is used for simulation test. The computer used in the simulation experiment is as follows: The processor is Intel(R) Core(TM) i3-3217U CPU @ 1.80GHz; The operating system is Windows 7 64-bit operating system.

In this 3D rugged terrain of the task environment, RRT can realize the path planning from the starting point to the target point. Fig. 4 and Fig. 5 respectively show the path in contour map and 3D map.

The path planning took 3.28s, and the length of path is 123.7. The path length is less than L_{max}. There was no sharp turn in the path. As shown in Fig. 4 and Fig. 5, RRT extends primarily in low elevation areas, avoiding peaks and rough terrain near its starting position. In most cases, the path passes through valleys and saddle areas. The low altitude area can guarantee the safety of the robot and help to make use of the shielding effect of surrounding terrain, so that the robot can reach the target position safely.

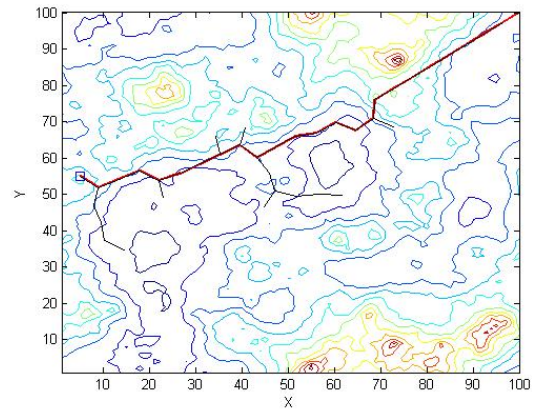

Figure 4: Path planning results (contour map).

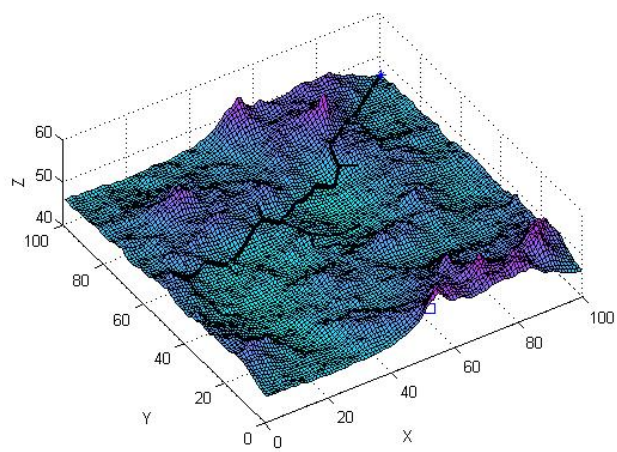

Figure 5: 3D path result.

In order to avoid the uncertainty effect of single test, the average value of 50 tests was taken. The average time of path planning is 3.12 second, and the average length of path is 124.5. The path obtained from each experiment can avoid peaks and rough terrain.

5 CONCLUSIONS

According to the special requirements of path planning in 3D environment, an improved RRT algorithm is proposed. Firstly, the robot's turning radius, range, pitch angle and other constraints are considered. Then, the dynamic p_g value method and the new node acceptance method are proposed. The

effectiveness and advance of the improved RRT algorithm are verified by simulation experiment. RRT extends primarily in low elevation areas, avoiding peaks and rough terrain near its starting position. In most cases, the path passes through valleys and saddle areas. The improved RRT algorithm realizes the path planning of 3D environment and can be widely used in the path planning of robot and UAV.

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